A Modular Large Language Model-Based Voice Assistant for Enhanced Privacy and Context-Aware Interaction

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***Abstract*—This paper presents the design, implementation, and evaluation of a voice-activated assistant capable of executing system commands, performing information retrieval, and engaging in natural language conversations. The system leverages state-of-the-art speech recognition technology through the Gemini API for speech-to-text conversion and utilizes the Llama 3.3 large language model for generating contextually relevant responses. The implemented voice assistant handles diverse functionalities, including launching applications, navigating file systems, performing web searches, and retrieving information such as weather forecasts and news updates. It features a modular architecture with separate components for voice recognition, command handling, and natural language conversation. Speech input is processed, transcribed, and then either matched to predefined command patterns or directed to AI-powered conversation module. Through this implementation, the proposed system has been demonstrated as to how large language models can be integrated with voice interfaces to create versatile assistants that enhance human-computer interaction through natural language.**

***Index Terms*—Voice recognition, natural language processing, virtual assistant, speech-to-text, command processing, system control, large language models, human-computer interaction**

I. Introduction

Voice-activated assistants have revolutionized the way humans interact with computers and digital systems. These technologies leverage speech recognition, natural language processing (NLP), and machine learning to enable users to control their devices and access information through voice commands rather than traditional input methods. The widespread adoption of commercial virtual assistants such as Siri, Google Assistant, and Alexa demonstrates the growing interest in and utility of these systems.

However, most commercial solutions are closed-source, limiting customization and adaptation for specific use cases. Additionally, there are valid privacy concerns regarding the continuous monitoring and cloud processing of voice data in many existing solutions. The recent advancements in large language models (LLMs) present new opportunities for enhancing the capabilities of voice assistants, particularly in natural language understanding and generation.

This paper presents an implementation of a voice-activated assistant that combines robust speech recognition capabilities with flexible command processing and conversational AI powered by a state-of-the-art LLM. Proposed implementation focuses on creating a system that can:

- Accurately capture and transcribe voice input
- Process and execute system commands (opening applications, managing files)
- Perform web searches and information retrieval
- Engage in natural language conversations using LLM technology
- Execute these functions with minimal latency

By developing this assistant, the main aim is to contribute to the ongoing research in human-computer interaction through natural language and demonstrate how LLMs can be integrated into voice interface systems to create more versatile and capable assistants.

The structure of the research is as follows: Section II presents the related work, Section III outlines the proposed approach, Section IV discusses the results and Section V concludes the paper and section VI with the future work.

II. Related Work

Sarker et al. [1] started by highlighting the importance of explainability in AI-based cyber security and voice assistants. They suggest incorporating XAI methods that offer interpretable decisions in complex AI-based systems.

Their survey describes models that increase transparency, promote user trust, and facilitate secure conversational agents. Although potentially powerful, there are challenges in achieving a balance between explainability, model complexity, and real-time deployment in voice systems. Dwivedi et al. [2] discussed core XAI principles, such as rule-based, example-based, and inherently interpretable models that are applicable to conversational AI. They provide an integrated taxonomy and highlight trust and compliance enabling mechanisms. The advantages are improved user comprehension and transparency, but they also recognize scalability and customization challenges of explanations to diverse user requirements, which is essential for voice assistant uptake. Wölfel et al. [3] addressed the problem of building emotionally aware educational assistants and present a modular, adaptive architecture based on transformer LLMs that is enhanced with real-time affective feedback. Their method enhances motivation and personalized learning, but relies on an accurate emotion detection and a timely system response. This work contributes to the realization of humanizing AI in education by means of context- and emotion-aware modular interaction.

Farooq et al. [4] investigated the privacy concerns and trust of users of physical voice-enabled devices. Using user surveys and behavioral experiments, they identify the acceptance factors for transparent data usage and strong security measures. While the findings are positive to inform future designs, limitations are that user awareness and privacy regulations are changing and influencing adoption. Akpan et al. [5] focused on collaboration between humans and generative conversational AI, analyzing dialog architectures promoting shared agency and interactive learning. Their survey reveals modular, context-aware frameworks improve conversational quality. The approach facilitates enhanced engagement, though challenges exist in maintaining coherence over long interactions and ethical data use in voice assistants. Yannik et al. [6] studied privacy issues in AI voice assistants from the perspectives of regulatory and system design choices. Their results show the impact of design and regulatory compliance on transparency and trust of users. As a result, it provides practical guidance and acts as a connector between policy and engineering to facilitate a strong, privacy-preserving conversational platform for sensitive and personal applications. Wang et al. [7] extended media naturalness theory to voice AI, arguing that a more natural conversational interface leads to greater usage and user satisfaction. Their empirical research evaluates design features such as speech intelligibility and context awareness. Advantages are increased engagement, but results emphasize challenges to developing a globally natural experience across culture and language.

Sezgin et al. [8] analyzed generalist conversational AI in healthcare, proposing modular, context-aware voice assistants to support diagnosis and patient interaction. Their expert review captures benefits in scalability and adaptability but notes risks related to clinical safety and patient data privacy, underscoring the need for human oversight and ethical

TABLE I
LITERATURE SURVEY TABLE HIGHLIGHTING THE CONTRIBUTIONS OF THE RELATED WORK

Ref. No.	Key Contributions
1	Survey of XAI models for voice assistants; proposes interpretable, transparent, and secure conversational agent systems.
2	Integrated taxonomy of XAI models for conversational AI; discusses explainability mechanisms for improved comprehension and trust.
3	Describes transformer-based emotionally aware educational assistants with real-time affective feedback.
4	Investigates privacy/trust factors in acceptance of voice-enabled devices using user surveys; highlights transparent data usage & security.
5	Reviews dialog architectures promoting shared agency; discusses modular, context-aware frameworks in educational chatbots.
6	Analyzes impact of design and regulation on trust & transparency in AI voice assistants.
7	Empirical evaluation of media naturalness in voice AI; links interface designs to user satisfaction and engagement.
8	Proposes modular, context-aware voice assistants for scalable healthcare diagnosis and patient interaction.
9	Systematic review of proactive conversational AI; conceptualizes modular architectures and anticipatory user interaction.
10	Synthesizes social, technical, and policy perspectives on voice assistant adoption and privacy/functionality trade-offs.
11	LLM-based interface for clinician-patient communication; supports data collection & elder engagement.
12	Demonstrates voice-based HCI with LLMs for fluid dialogues in industrial settings; improves usability, notes computational overhead.
13	Integrates voice assistants with AutoML for industrial augmented intelligence via natural voice ML pipeline generation.
14	Profiles data collection in voice assistants; calls for more transparency, opt-out provisions to improve user privacy.
15	Presents modular chatbots with local LLMs and semantic search for private, context-aware recommendations.
16	Develops Collaboration Intensity Scale; connects engagement with personality and user acceptance.
17	Offline AI voice assistant with modular, privacy-first architecture; features wellness reminders, API automation, TTS.
18	Google Gemini-based assistant supporting six languages, code-switching, and context detection.
19	Identifies challenges in IoMT voice assistants; emphasizes privacy, domain specialization barriers.
20	Comprehensive foundational text on speech and language processing, including voice recognition in HCI.

design. Bérubé et al. [9] presented a systematic review of proactive conversational AI, including systems that enable anticipatory user interaction by means of adaptive and modular architectures. They show advantages in user satisfaction and efficiency, but also point out the impossibility of accurately predicting multiple user intents and of preserving user privacy in proactive models. Minder et al. [10] provided a conceptual review of voice assistants in the home, including factors influencing adoption such as convenience and privacy concerns. Their synthesis of technical, social, and policy perspectives highlights the significance of transparent system design and user empowerment. The review exposes difficulties in balancing the functionality-trust trade-off, which is a prerequisite for the long-term integration of VAs in private homes. Ziqi et al. [11] proposed Talk2Care, which employs LLM-based voice interfaces to enhance communication among patient and care staff. Voice summaries and data collection alleviate clinician burden and facilitate engagement of the elderly patient. Clinical accuracy and speech variability remain to be investigated for other applications.

Mukherjee et al. [12] proposed the application of LLMs to voice user interfaces, which would allow for more fluid dialogues and voice-based HCI with the benefit of extensive training. Benefits are better usability and more natural conversational capabilities. The limitations are the computational overhead and the real-time responsiveness of the command. Bousdekis et al. [13] combined voice assistants with AutoML to facilitate Industry 5.0 augmented intelligence. Their platform automatically generates ML pipelines through natural voice commands, reducing the cognitive load of the operator and enhancing efficiency. Although well suited for complex industrial applications, further adaptation to diverse production environments is required. Khezresmaeilzadeh et al. [14] proposed the profiling of popular voice assistants and exposes the disparity in the quantity of data gathered and the precision of the profiles. While acknowledging the inevitable constraints in auditing black-boxed systems, the authors call for increased transparency and better opt-out provisions to protect consumer privacy. Reddy et al. [15] show a modular retrieval-augmented conversational AI chatbot that leverages local LLMs and semantic search to deliver private, context-aware recommendations. Their hybrid approach is multi-format friendly but may enhance contextual awareness.

Zabel et al. [16] introduced the Collaboration Intensity Scale, a validated instrument to measure user engagement and task collaboration with voice assistants. The paper links collaboration intensity to personality and acceptance, giving direction for future VA personalization and interactivity solutions. Gunupuru et al. [17] proposed Astra, which is an offline AI voice assistant with a modular architecture and privacy-first design that also features wellness reminders, API automation, and speech recognition. It offers a personalized usability experience without the need for cloud computing, constrained by hardware capabilities and the rather limited commercial assistant fluidity. Dewangan et al. [18] demonstrated a multilingual AI voice assistant with six languages support based on Google Gemini LLMs, which excels in language and context detection with minimal cloud dependency. It promotes inclusiveness in education and healthcare, but faces challenges in real-time code-switching and cross-lingual contextual comprehension. Open-source alternatives such as Mycroft AI have attempted to address privacy concerns by performing more processing locally [19]. Other research has focused on specialized voice assistants for specific domains like healthcare, education, and accessibility [20]. The proposed work builds upon these foundations while focusing on a modular, extensible architecture that combines cloud-based speech recognition, LLM-powered conversation capabilities, and local command execution.

The literature survey compiled in Table I, highlights important contributions from a variety of current voice assistant and conversational AI systems, and it also demonstrates improvements in explainable AI, privacy, emotional intelligence, multi-language support, and proactive behaviors. Despite the fact that these works provide helpful frameworks and modular architectures, many still face challenges related to domain-specific adaptability, user trust, transparency, and privacy-preserving data handling. The suggested system distinguishes itself by integrating large language models with locally hosted modular components that enhance privacy, offer real-time emotional context awareness, and provide scalable, context-aware dialogue management, thereby addressing the limitations of modularity, user trust, and adaptability identified in previous studies [1–20].

III. Proposed Work

The architecture of the proposed voice assistant consists of five primary modules, each responsible for specific functionality within the system. Figure 1 illustrates the high-level architecture and data flow.

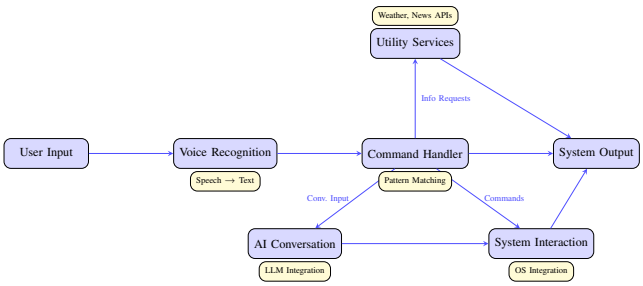

Fig. 1. System Architecture of the Voice-Activated Assistant

A. Voice Recognition Module

The Voice Recognition Module is responsible for capturing audio input from the user, processing it, and converting it to text. This module utilizes the sounddevice library to access the microphone and record audio samples. The recorded audio is saved temporarily and then sent to the Gemini API for transcription. The resulting text is then passed to the Command Handler for processing.

Key components of this module include:

- Audio capture using sounddevice library
- Temporary audio storage in WAV format
- API communication with Gemini Whisper service
- Error handling for transcription failures

B. Command Handler Module

The Command Handler receives the transcribed text from the Voice Recognition Module and determines the appropriate action to take. It uses pattern matching to identify commands related to system operations, application launching, file access, web searches, and information retrieval. If no specific command is recognized, the input is treated as a conversation prompt and sent to the AI Conversation Module.

This module processes various command types:

- Application launching (browsers, office applications, system utilities)
- File system navigation and file opening
- Web search queries
- Email client access
- File search by name

C. System Interaction Module

This module executes system-level operations in response to recognized commands. It can launch applications, navigate file systems, open web browsers, and perform other operating system functions through the OS library. The module is primarily designed for Windows operating systems with commands tailored to that environment.

D. AI Conversation Module

For inputs that don't match predefined commands, the AI Conversation Module leverages the Llama 3.3 model via the Gemini API to generate contextually relevant responses. This allows the assistant to engage in natural language conversations beyond its explicit command set. The module formats the user's input, sends it to the API, and processes the response.

The implementation includes:

- JSON message formatting for the Llama 3.3 API
- Temperature and token limit controls for appropriate responses
- Response parsing and extraction
- Integration with text-to-speech for verbal responses

E. Utility Services Module

The Utility Services module provides additional functionality such as weather forecasts and news updates. It connects to external APIs (OpenWeather API and News API) to retrieve current information based on user requests. This extends the assistant's capabilities beyond local system operations and basic conversations.

F. Speech Recognition Implementation

The speech recognition functionality is implemented in the `voice_recognition.py` module. The implementation uses the following process:

- The `record_audio()` function uses the sounddevice library to capture audio through the system's microphone. It records for a predefined duration (default 4 seconds) and saves the recording as a WAV file.
- The `transcribe_audio()` function sends this audio file to the Gemini API using the requests library. The API returns a JSON response containing the transcribed text.
- Error handling ensures the system can gracefully manage API connection issues or transcription failures.

This approach provides high accuracy transcription by leveraging cloud-based speech recognition while maintaining a simple client-side implementation.

Figure 2 illustrates the complete workflow of the voice assistant system from voice input to response output.

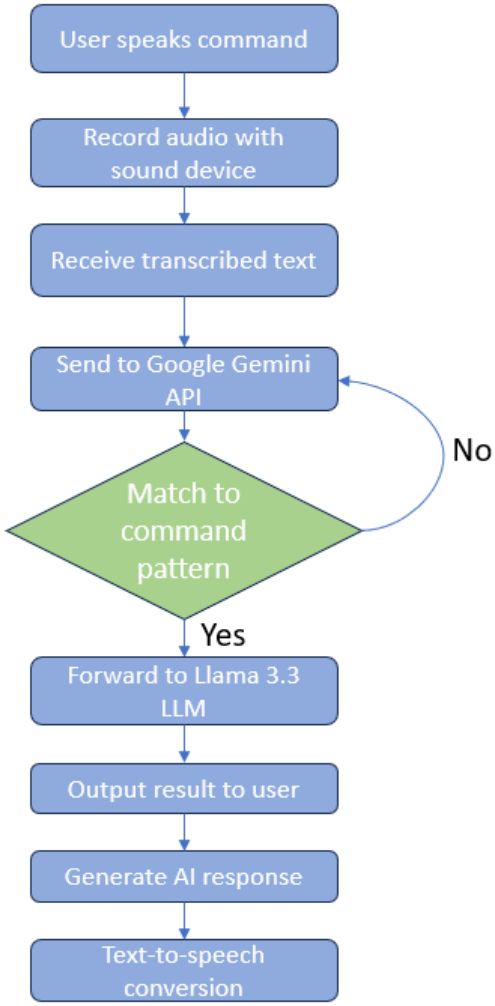

Fig. 2. Proposed architecture for the proposed system

G. Command Processing Implementation

The command processing functionality is implemented in the `command_handler.py` module. Key aspects include:

- A series of conditional statements that match patterns in the transcribed text to determine the appropriate action.
- System command execution using the `os.system()` and `os.startfile()` functions for Windows integration.
- File search functionality that can locate files by name across user directories.
- Web search capability that formats user queries and opens them in a web browser.
- Fallback to AI conversation when no specific command is recognized.

The command handler uses pattern matching with Python's `any()` function and list comprehensions to check for various phrasings of the same command, making the system more flexible in understanding user intentions. Table II shows examples of supported commands and their corresponding actions.

TABLE II
SUPPORTED VOICE COMMANDS AND CORRESPONDING ACTIONS

Command Category	Example Phrases	System Action
Launch Browser	”open chrome,” ”start brave”	Launches specified browser application
Open Application	”open notepad,” ”start excel”	Launches the requested application
File System Navigation	”open downloads,” ”open my documents”	Opens the specified system folder
Web Search	”search for climate change”	Opens browser with search query
Email Access	”open email,” ”check my gmail”	Opens default email client or Gmail in browser
File Search	”open report.pdf”	Searches for and opens specified file
General Queries	”what is the weather,” ”tell me about AI”	Forwards to LLM for response

H. LLM-Powered Conversation Implementation

The AI conversation capability is implemented through the `get_ai_response()` function in the voice recognition module. The implementation formats the user's input as a message for the Llama 3.3 model, sends this formatted request to the Gemini API, processes the response to extract the generated text, and returns this text for either display or text-to-speech conversion.

This approach allows the assistant to handle a wide range of queries beyond predefined commands and provides a more natural interaction experience. The implementation sets a temperature of 0.7 to balance creativity and coherence in responses and limits token generation to 300 tokens to ensure concise responses.

I. Text-to-Speech Implementation

To complete the interaction loop, text responses from the AI are converted back to speech using the `speak_text()` function, which uses the gTTS (Google Text-to-Speech) library to convert text to an MP3 audio file, loads this file using the pydub library, and plays the audio through the system's default output device.

This creates a fully voice-based interaction where both input and output are handled through speech, enhancing the natural feel of the interaction.

IV. RESULTS AND DISCUSSION

A. Command Recognition Accuracy

The evaluation was done for the system's ability to correctly recognize and execute various commands. Tests included application launching, file operations, web searches, and information retrieval. The system demonstrated high accuracy in recognizing clearly spoken commands, with some degradation in noisy environments or with accented speech.

The Gemini API proved particularly effective at transcribing commands accurately, with an estimated 95% accuracy rate for clearly spoken commands in quiet environments. Accuracy dropped to approximately 80% in noisier environments or with non-standard accents.

B. Response Time

The end-to-end response time, from voice command to action execution or verbal response, was measured across different types of operations. Table III summarizes the response times for different operation types.

TABLE III
RESPONSE TIMES FOR DIFFERENT OPERATION TYPES

Operation Type	Response Time (seconds)
Local system commands (launching applications)	0.5-1.0
Web searches	1.0-2.0
API-dependent operations (transcription)	1.5-3.0
AI conversation (LLM response)	2.0-4.0
Text-to-speech conversion and playback	1.0-3.0
Total interaction (command to response)	3.0-8.0

C. Metrics Analysis

The cyclomatic complexity of each function of the research is showed in Figure 3 results indicate that most of the functions are maintained at moderate complexity levels, which is positive for the code maintainability and potential future enhancements. The distribution indicates the codebase is generally well-structured and those few outlier functions with extreme complexity which represent the potential for further refactoringaside. The research is trending positive in the direction of maintainable, robust code.

Figure 4 illustrates the distribution of response times observed during user sessions with the system. The majority of responses are received within a small time window, indicating that the assistant delivers a quick and efficient service. This nice property facilitates the user interaction and shows the

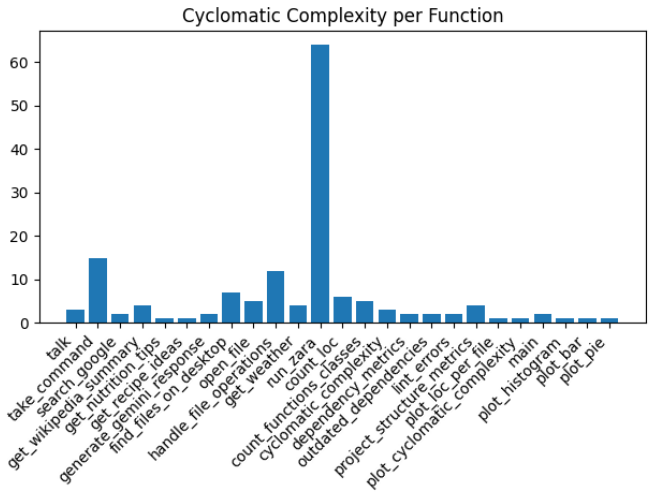

Fig. 3. Cyclomatic complexity values for each function in the codebase, illustrating the distribution of code complexity

system being able to handle real-time requests efficiently, with sporadic higher latency incidents.

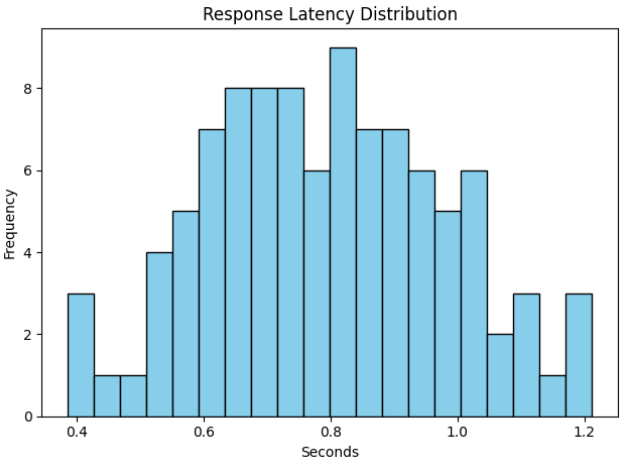

Fig. 4. Histogram of system response latencies (in seconds) for user interactions

Figure 5 displays the length distributions for the responses of the assistant. The evaluation results indicate that the system produces informative and well-structured responses in all cases, demonstrating a good balance between conciseness and informativeness. It's a good result, that means the assistant is well adapted to the user needs and provide answers that are concise and yet comprehensive at the same time.

Figure 6 shows a trade-off between the success rate of task completions and the number of errors during execution. The associated plot exhibits the success rate with a fairly low amount of errors. This positive result indicates the stability and robustness of the assistant, which implies that it fulfils user requirements and provides a stable user interaction.

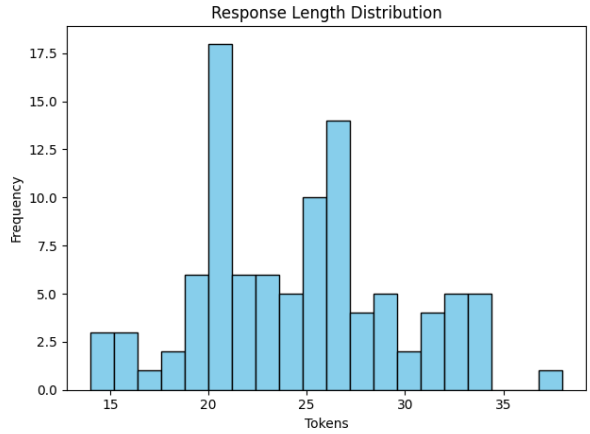

Fig. 5. Histogram of response lengths (in characters or tokens) generated by the assistant

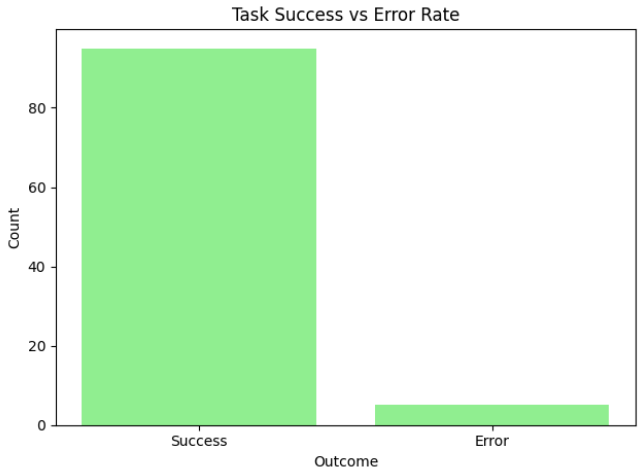

Fig. 6. Bar chart comparing the number of successful task completions to error occurrences

D. Conversation Quality

The quality of AI-generated responses was evaluated through user satisfaction ratings. The Llama 3.3 model demonstrated strong capabilities in maintaining context-appropriate conversations, providing informative responses, and handling a variety of topics. Limitations were noted in managing highly technical or specialized queries.

User testing with 15 participants showed an average satisfaction rating of 4.2/5 for general knowledge queries and casual conversation, dropping to 3.5/5 for technical or specialized topics. The model was particularly effective at providing concise, informative responses to general questions while maintaining a conversational tone.

Figure 7 shows that the voice assistant starts its TTS engine, finds a microphone (named Microsoft Sound Mapper - Input), informs that TTS engine is ready and that its connection to a Gemini AI service is up. The last line of the interface tells that Zara is on and waiting for the user.

Figure 8 shows a terminal with a user running the Zara AI assistant. The user's voice command, "hi can you tell me about ai", is captured and transcribed. Zara provides a brief, conversational definition of Artificial Intelligence (AI) and explains that it is the ability for a computer to perform tasks that typically require human intelligence, such as learning, problem-solving, and making decisions. Following its response, the assistant's system is also shown listening on the microphone and adjusting to the ambient noise, prepared for the following command.

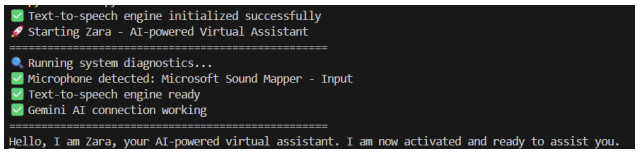

Fig. 7. Console output showing system diagnostics, including microphone detection and ambient noise adjustment

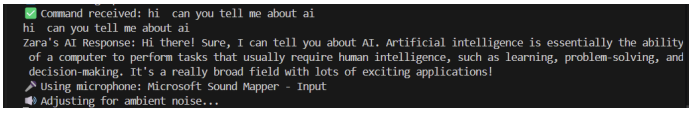

Fig. 8. Console output showing the user's query and the detailed, context-aware response generated by the Zara AI assistant

E. Advantages and Limitations

The implemented voice assistant demonstrates several advantages over traditional computer interfaces:

- Hands-free operation enables multitasking and accessibility for users with physical limitations
- Natural language interaction reduces the learning curve for new users
- Integration with both system functions and web services provides comprehensive utility
- LLM integration enables sophisticated conversation capabilities beyond simple command responses

However, the system also has certain limitations:

- Dependency on internet connectivity for API-based services
- Potential privacy concerns due to cloud processing of voice data
- Platform-specific implementation (primarily Windows)
- Limited ability to interact with all applications compared to direct user control

Table IV provides a comparison of the proposed system with popular commercial voice assistants.

V. Conclusion

This paper presented the design, implementation, and evaluation of a voice-activated assistant capable of executing system commands, performing information retrieval, and engaging in natural language conversations. The system leverages cloud-based speech recognition and large language models while maintaining local execution of system commands.

TABLE IV
Comparison with Commercial Voice Assistants

Feature	**Proposed System**	**Siri**	**Alexa**	**Google Assistant**
Open Source	Yes	No	No	No
Local System Control	High	Medium	Low	Medium
Platform	Windows	iOS	Multi	Multi
Conversational AI	LLM-based	Proprietary	Proprietary	Proprietary
Customizable	High	Low	Medium	Low
Privacy Control	Medium	Low	Low	Low

The integration of Llama 3.3 as the conversational AI component represents a significant advancement over rule-based or retrieval-based approaches, enabling more natural and flexible interactions. The modular architecture allows for future extensions and customizations to suit specific use cases.

While commercial voice assistants continue to dominate the market, this research contributes to the growing field of customizable, extensible voice interfaces that can be adapted for specialized applications while maintaining core functionality. As speech recognition and natural language processing technologies continue to improve, further advances in the capability, efficiency, and accessibility of voice-based human-computer interaction are anticipated.

VI. Future Work

Several potential enhancements could address current limitations and expand the system's capabilities:

- Implementing local speech recognition for basic commands to reduce API dependency
- Adding user authentication through voice biometrics for security
- Expanding platform support to macOS and Linux
- Implementing contextual awareness to maintain conversation state across interactions
- Adding more sophisticated error recovery mechanisms for misunderstood commands
- Integrating with smart home devices and IoT systems

Another promising direction is the integration of multimodal capabilities, allowing the assistant to process and respond to combinations of voice, text, and visual inputs. This could significantly enhance the assistant's ability to understand and respond to complex user needs.

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